

# Resources

**STAT 109 — Summer Session 1, 2026 (online, async)**

# Resources

Everything you read and practice lives on this website. Everything you submit for a grade lives in [Canvas](#).

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## Start here

| Resource    | Link                     | When to use                        |
|-------------|--------------------------|------------------------------------|
| Course path | <a href="#">Schedule</a> | readings, labs, quizzes, projects  |
| Syllabus    | <a href="#">Syllabus</a> | Policies, grading, AI expectations |

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## Reference sheets

| Sheet       | Link                        | Purpose                                                    |
|-------------|-----------------------------|------------------------------------------------------------|
| Methods map | <a href="#">Methods</a>     | When to use which test or summary; study-design vocabulary |
| R reference | <a href="#">R reference</a> | Code patterns and syntax                                   |

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## R and notebooks

This async course runs R in Google Colab. Lab and project pages link to notebooks on the public GitHub repo; open a link and Save a copy to Drive before editing.

| Step | What to do                                                                                       |
|------|--------------------------------------------------------------------------------------------------|
| 1    | Open the Colab link from a <a href="#">demo lab</a> or <a href="#">exercise lab</a>              |
| 2    | <b>File</b> → <b>Save a copy in Drive</b> (or download <code>.ipynb</code> and upload to Canvas) |
| 3    | Run cells top to bottom; fix errors before submitting                                            |
| 4    | Submit the completed <code>.ipynb</code> in Canvas                                               |

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**Colab home:** [colab.research.google.com](https://colab.research.google.com) (remember to change runtime to R as Python 3 is the default for new notebooks)

**Colab tutorial:** [Google Colab overview](#) — running cells, saving a copy to Drive, editing notebooks. Keep in mind that Colab can be used for multiple programming languages but the basic use apply to how we'll use it to program in the R language.

## Optional: R on your own computer

You do not need a local install for this course. If you prefer RStudio or Jupyter on your machine:

- **R:** [CRAN](#) · [Windows/Mac/Linux installers](#)
- **RStudio (Posit):** [RStudio Desktop](#) — install R first
- **R in Jupyter:** [IRkernel setup](#)

R is also installed on campus computers ([ROSE](#) may have free used laptops if you need hardware).

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## Quarto and reproducible reports

Projects and some readings use **Quarto** (`.qmd`) — text + code that renders to HTML or PDF. Optional if you want reproducible write-ups beyond plain notebooks:

- [Quarto documentation](#)
- [Posit cheatsheets](#) — R, `{tidyverse}`, `{ggplot2}`, and more

Render locally: `quarto render your-file.qmd` (requires [Quarto](#) and R).

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## Textbook (optional)

[Introductory Statistics for the Life and Biomedical Sciences](#) by Julie Vu and David Harrington — free PDF at [OpenIntro](#) or [Leanpub](#). Module reads on this site have all the essential topics but it can be useful to see alternative explanations in someone else's words.

**Note:** The textbook ships with [companion R labs](#) written in base R. This course uses Google Colab, `{tidyverse}`, and `{ggplot2}` in our own lab notebooks.

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## Practice quiz banks

Open-ended practice questions for each quiz. Canvas quizzes use a random autograded subset.

| Module | Quiz A                           | Quiz B                         | Synthesis                 |
|--------|----------------------------------|--------------------------------|---------------------------|
| 0      | <a href="#">Start Here Check</a> | <a href="#">Question Ideas</a> | —                         |
| 1      | <a href="#">Def</a>              | <a href="#">Concept</a>        | <a href="#">Synthesis</a> |
| 2      | <a href="#">Def</a>              | <a href="#">Concept</a>        | <a href="#">Synthesis</a> |
| 3      | <a href="#">Def</a>              | <a href="#">Concept</a>        | <a href="#">Synthesis</a> |
| 4      | <a href="#">Def</a>              | <a href="#">Concept</a>        | <a href="#">Synthesis</a> |
| 5      | <a href="#">Def</a>              | <a href="#">Concept</a>        | <a href="#">Synthesis</a> |

**Image-based graph items** are in the **Module 1** synthesis bank (binomial null plots) and **Module 3** synthesis bank (histograms, boxplots, scatterplots, stacked bars). The same figures are grouped on one page: [Graph Interpretation Pack](#).

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## Optional Supplemental Materials by module

| Module   | Textbook (Vu & Harrington)                                                                                                | Videos & tutorials                                                                                                                                      |
|----------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>0</b> | <a href="#">Preface / book home</a>                                                                                       | <a href="#">Colab tutorial</a> · <a href="#">Posit: Getting Started with R</a> · <a href="#">Navarro: R for beginners</a>                               |
| <b>1</b> | <a href="#">Ch 2 Probability</a> · <a href="#">Ch 3 Distributions (§3.2 binomial)</a>                                     | <a href="#">Khan: binomial distribution</a> · <a href="#">Khan: visualizing binomial</a>                                                                |
| <b>2</b> | <a href="#">Ch 3.3 Normal</a> · <a href="#">Ch 4 Foundations (§4.1–4.2)</a>                                               | <a href="#">Khan: confidence intervals</a> · <a href="#">Khan: CI simulation</a> · <a href="#">Mine Çetinkaya-Rundel (search: confidence intervals)</a> |
| <b>3</b> | <a href="#">Ch 1 Intro to data (§1.4–1.7)</a>                                                                             | <a href="#">R4DS: Data visualization</a> · <a href="#">Posit: ggplot2 video</a>                                                                         |
| <b>4</b> | <a href="#">Ch 5 Inference for numerical data (§5.3)</a> · <a href="#">Ch 8 Inference for categorical data (§8.1–8.2)</a> | <a href="#">R4DS: Data transformation</a> · <a href="#">Andrew Heiss (workflow, comparing groups)</a>                                                   |
| <b>5</b> | <a href="#">Ch 6 Simple linear regression</a>                                                                             | <a href="#">Khan: regression line</a> · <a href="#">Penn State: prediction intervals</a> · <a href="#">Regression shuffle applet</a>                    |

## Site-wide companions

| Resource              | Link                                                                                | Best for                                                        |
|-----------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| R for Data Science    | <a href="https://r4ds.hadley.nz">r4ds.hadley.nz</a>                                 | <code>{tidyverse}</code> + <code>{ggplot2}</code> (Modules 3–5) |
| Posit cheatsheets     | <a href="https://posit.co/resources/cheatsheets">posit.co/resources/cheatsheets</a> | Building your <b>R reference</b>                                |
| Posit videos          | <a href="https://posit.co/resources/videos">posit.co/resources/videos</a>           | Beginner R workflow                                             |
| Mine Çetinkaya-Rundel | <a href="https://youtube.com/@minecr">youtube.com/@minecr</a>                       | Statistics-first explanations                                   |

## Getting help

- **Email:** [rho3@humboldt.edu](mailto:rho3@humboldt.edu) (Cal Poly Humboldt account; allow one business day during the session)
- **Office hours:** By appointment and in Canvas Discussion
- **Canvas:** Announcements, grades, and module discussions

## Campus support

- **Computer labs:** R installed on campus machines
- **ROSE:** [Reusable Office Supply Exchange](#) — free used computers when available
- **SDRC:** [Student Disability Resource Center](#) — accommodations (email instructor your letter of accommodation)